

23CS2014 Database Systems Lab

Ex. No. 1(a) – Creating and Managing Tables

Aim

To execute DDL commands and get the desired output.

Description

DDL refers to "Data Definition Language", a subset of SQL statements that change the structure of the database schema in some way, typically by creating, deleting, or modifying schema objects such as databases, tables, and views. Most Impala DDL statements start with the keywords CREATE , DROP , or ALTER .

In this schema, we have four main tables: User, Event, Venue, and Ticket.

User table:

Column	Data Type
UserID	NUMBER(10)
Name	VARCHAR2(255)
Email	VARCHAR2(255)
Password	VARCHAR2(255)
Phone	VARCHAR2(20)

Event table:

Column	Data Type
EventID	NUMBER(10)
Name	VARCHAR2(255)
Date	DATE
Time	TIMESTAMP
VenueID	NUMBER(10) (Foreign key)
Description	VARCHAR2(500)

Venue table:

Column	Data Type
VenueID	NUMBER(10)
Name	VARCHAR2(255)
Address	VARCHAR2(255)
City	VARCHAR2(255)
State	VARCHAR2(255)
Country	VARCHAR2(255)

Ticket table:

Column	Data Type
--------	-----------

23CS2014 Database Systems Lab

TicketID	NUMBER(10)
EventID	NUMBER(10) (Foreign key)
UserID	NUMBER(10)
SeatNumber	VARCHAR2(20)
Price	NUMBER(10, 2)
Status	VARCHAR2(50)

Queries(DDL Commands)

1. Create User, Event, Venue, and ticket tables based on the given schema.

```
CREATE TABLE VenueURK24CS1006 (  
    VenueID INT PRIMARY KEY,  
    Name VARCHAR(255),  
    Address VARCHAR(255),  
    City VARCHAR(255),  
    State VARCHAR(255),  
    Country VARCHAR(255)  
);
```

```
CREATE TABLE UsersURK24CS1006 (  
    UserID INT PRIMARY KEY,  
    Name VARCHAR(255),  
    Email VARCHAR(255),  
    Password VARCHAR(255),  
    Phone VARCHAR(20)  
);
```

```
CREATE TABLE EventURK24CS1006 (  
    EventID INT PRIMARY KEY,  
    Name VARCHAR(255),
```

23CS2014 Database Systems Lab

```
EventDate DATE,  
  
    EventTime TIMESTAMP,  
  
    VenueID INT,  
  
    Description VARCHAR(500)  
  
);  
  
CREATE TABLE TicketURK24CS1006 (  
  
    TicketID INT PRIMARY KEY,  
  
    EventID INT,  
  
    UserID INT,  
  
    SeatNumber VARCHAR(20),  
  
    Price DECIMAL(10,2),  
  
    Status VARCHAR(50)  
  
);
```

```
mysql> use dbmslab  
Database changed  
mysql> CREATE TABLE VenueURK24CS1006 (VenueID INT PRIMARY KEY, Name VARCHAR(255), Address VARCHAR(255), City VARCHAR(255), State VARCHAR(255), Country VARCHAR(255));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> CREATE TABLE UsersURK24CS1006 (UserID INT PRIMARY KEY, Name VARCHAR(255), Email VARCHAR(255), Password VARCHAR(255), Phone VARCHAR(20));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> CREATE TABLE EventURK24CS1006 (EventID INT PRIMARY KEY, Name VARCHAR(255), EventDate DATE, EventTime TIMESTAMP, VenueID INT, Description VARCHAR(500));  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> CREATE TABLE TicketURK24CS1006 (TicketID INT PRIMARY KEY, EventID INT, UserID INT, SeatNumber VARCHAR(20), Price DECIMAL(10,2), Status VARCHAR(50));  
Query OK, 0 rows affected (0.01 sec)
```

23CS2014 Database Systems Lab

2. Describe the tables.

```
DESC UsersURK24CS1006;
```

```
DESC EventURK24CS1006;
```

```
DESC VenueURK24CS1006;
```

```
DESC TicketURK24CS1006;
```

```
mysql> DESC UsersURK24CS1006;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| UserID | int | NO | PRI | NULL |  |
| Name | varchar(255) | YES |  | NULL |  |
| Email | varchar(255) | YES |  | NULL |  |
| Password | varchar(255) | YES |  | NULL |  |
| Phone | varchar(20) | YES |  | NULL |  |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> DESC EventURK24CS1006;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| EventID | int | NO | PRI | NULL |  |
| Name | varchar(255) | YES |  | NULL |  |
| EventDate | date | YES |  | NULL |  |
| EventTime | timestamp | YES |  | NULL |  |
| VenueID | int | YES |  | NULL |  |
| Description | varchar(500) | YES |  | NULL |  |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> DESC VenueURK24CS1006;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| VenueID | int | NO | PRI | NULL |  |
| Name | varchar(255) | YES |  | NULL |  |
| Address | varchar(255) | YES |  | NULL |  |
| City | varchar(255) | YES |  | NULL |  |
| State | varchar(255) | YES |  | NULL |  |
| Country | varchar(255) | YES |  | NULL |  |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> DESC TicketURK24CS1006;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| TicketID | int | NO | PRI | NULL |  |
| EventID | int | YES |  | NULL |  |
| UserID | int | YES |  | NULL |  |
| SeatNumber | varchar(20) | YES |  | NULL |  |
| Price | decimal(10,2) | YES |  | NULL |  |
| Status | varchar(50) | YES |  | NULL |  |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> |
```

3. Alter the User table to add a new column Age.

23CS2014 Database Systems Lab

```
ALTER TABLE UsersURK24CS1006 ADD Age INT;
```

```
mysql> ALTER TABLE UsersURK24CS1006 ADD Age INT;  
Query OK, 0 rows affected (0.02 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
  
mysql> |
```

4. Drop the newly added column Age.

```
ALTER TABLE UsersURK24CS1006 DROP COLUMN Age;
```

```
mysql> ALTER TABLE UsersURK24CS1006 DROP COLUMN Age;  
Query OK, 0 rows affected (0.02 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
  
mysql> |
```

5. Rename the Venue table to Location

```
RENAME TABLE VenueURK24CS1006 TO LocationURK24CS1006;
```

```
mysql> RENAME TABLE VenueURK24CS1006 TO LocationURK24CS1006;  
Query OK, 0 rows affected (0.01 sec)  
  
mysql> |
```

6. Modify the size of the Event table's Description column to 1000.

```
ALTER TABLE EventURK24CS1006 MODIFY Description VARCHAR(1000);
```

```
mysql> ALTER TABLE EventURK24CS1006 MODIFY Description VARCHAR(1000);  
Query OK, 0 rows affected (0.01 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
  
mysql> |
```

7. Drop the SeatNumber column from the Ticket table

23CS2014 Database Systems Lab

```
ALTER TABLE TicketURK24CS1006 DROP COLUMN SeatNumber;
```

8. Add a unique constraint on the Email column in the User table

```
mysql> ALTER TABLE TicketURK24CS1006 DROP COLUMN SeatNumber;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> ALTER TABLE UsersURK24CS1006 ADD CONSTRAINT unique_email UNIQUE (Email);
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> |
```

```
ALTER TABLE UsersURK24CS1006 ADD CONSTRAINT unique_email UNIQUE
(Email);
```

9. Rename the UserID column in the User table to ID

```
ALTER TABLE UsersURK24CS1006 CHANGE UserID ID INT;
```

```
mysql> ALTER TABLE UsersURK24CS1006 CHANGE UserID ID INT;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> |
```

10. Modify the Ticket table to add a column named "Barcode" with a data type of VARCHAR2(50).

```
ALTER TABLE TicketURK24CS1006 ADD Barcode VARCHAR(50);
```

```
mysql> ALTER TABLE TicketURK24CS1006 ADD Barcode VARCHAR(50);
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> |
```

11. Modify the Name column in the Venue table to increase its maximum length to VARCHAR2(300)

23CS2014 Database Systems Lab

```
ALTER TABLE LocationURK24CS1006 MODIFY Name VARCHAR(300);
```

```
mysql> ALTER TABLE LocationURK24CS1006 MODIFY Name VARCHAR(300);  
Query OK, 0 rows affected (0.01 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

12. Add a foreign key constraint on the VenueID column in the Event table, referencing the Venue table.

```
ALTER TABLE EventURK24CS1006 ADD CONSTRAINT fk_event_venue FOREIGN  
KEY (VenueID) REFERENCES LocationURK24CS1006 (VenueID);
```

```
mysql> ALTER TABLE EventURK24CS1006 ADD CONSTRAINT fk_event_venue FOREIGN KEY (VenueID) REFERENCES LocationURK24CS1006(VenueID);  
Query OK, 0 rows affected (0.05 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
mysql> |
```

13. Add a CHECK constraint to check whether the UserID is between 101 and 105

```
ALTER TABLE UsersURK24CS1006 ADD CONSTRAINT check_userid CHECK (ID  
BETWEEN 101 AND 105);
```

```
mysql> ALTER TABLE UsersURK24CS1006 ADD CONSTRAINT check_userid CHECK (ID BETWEEN 101 AND 105);  
Query OK, 0 rows affected (0.04 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
mysql> |
```

14. Add a unique constraint to Phone column of the User table.

```
ALTER TABLE UsersURK24CS1006 ADD CONSTRAINT unique_phone UNIQUE  
(Phone);
```

```
mysql> ALTER TABLE UsersURK24CS1006 ADD CONSTRAINT unique_phone UNIQUE (Phone);  
Query OK, 0 rows affected (0.01 sec)  
Records: 0 Duplicates: 0 Warnings: 0  
mysql> |
```

15. Truncate the user table.

```
TRUNCATE TABLE UsersURK24CS1030;
```

23CS2014 Database Systems Lab

```
mysql> TRUNCATE TABLE UsersURK24CS1006;  
Query OK, 0 rows affected (0.03 sec)  
  
mysql> |
```

Result:

Thus, the various DDL commands to create, alter, drop, and manage the database tables were successfully executed, and the desired output was obtained.